Esha Pakalapati

647-619-6181 | esha.pakalapati@mail.utoronto.ca | linkedin.com/in/esha-pakalapati/ | github.com/rainyalgorithms

Education

University of Toronto Toronto, ON, Canada

Honours Bachelor of Science, Computer Science Specialist (Minor in Mathematics + PEY Co-op) *May 2027*

Relevant Coursework

Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Databases, Software Engineering, Information Security, OOP, Systems Programming (C/Unix), Python Programming, Software/Web Development, Computer Organization and Architecture, Probability and Statistics

Experience

Machine Learning Developer Intern May 2025 – Aug 2026

Airbus *Waterloo, ON*

- AI, ML & Optimization Chapter at Navblue, Airbus software subsidiary
- Accomplished 89% automation of a new ML-driven product pipeline by analyzing historical data, designing and training multiple PyTorch models on AWS SageMaker, and evaluating performance metrics.
- Improved training accuracy and efficiency by tuning hyperparameters (e.g., learning rate, patience), probing dataset structure, and optimizing input features using Python, Pandas, and Jupyter Notebooks.
- Enhanced model reproducibility and training scalability by debugging model scripts and streamlining preprocessing workflows in an Anaconda-based development environment.
- Won company-wide hackathon by building a prototype leveraging Gemini to automate internal tasks, demonstrating fast experimentation and integration skills.
- Developing a full-stack foosball tracking web app with other interns using FastAPI (backend), Vue (frontend), and AWS S3 for deployment and data storage.

Software Developer Jan 2025 – Apr 2025

Lunenfeld-Tanenbaum Research Institute (LTRI) and Ontario Institute for Cancer Research (OICR) *Toronto, ON*

- Built a full-stack, web-based cell segmentation enhancement tool to improve pathologists' workflow aiding in biomedical research.
- Accelerated segmentation reviews by allowing precise border edits and real-time feedback, improving diagnostic efficiency.
- Used Python, Flask, React, HTML/CSS, and JavaScript in an Agile environment with a 7-person team, integrating DeepCell, OpenCV, and NumPy.
- Implemented an AI-driven "segmentation refinement" feature that automatically suggests boundary corrections for missed or inaccurately segmented cells, reducing manual edits and enhancing accuracy.
- Working closely with Software Developer teams at *LTRI*, machine learning scientists, and pathologists to validate model accuracy, and coordinate tasks.
- Owned the end-to-end SDLC (requirements, design, development, testing, deployment) to ensure alignment with biomedical research needs.
- Improved UI/UX with a responsive React front end, enabling pathologists to highlight and adjust cell boundaries quickly.
- Maintained Git repositories with best practices for branching, pull requests, and merging, ensuring a smooth collaboration pipeline.

E-commerce Automation Developer Intern May 2024 – Aug 2024

Vinayak Indian Sarees - Etsy *Brampton, ON*

- Automated financial and inventory processes for a growing Etsy store.
- Reduced operational costs by 12% and boosted monthly profitability by 8% through faster data handling.
- Built a Python-based tool to manage records and track inventory, saving 14 hours/week in manual tasks.
- Resolved 85% of 5K+ customer inquiries within 2 hours, achieving a 95% satisfaction rate.
- Expanded to Shopify with $3K+ added monthly revenue, driving 35% growth and onboarding 100+ new customers monthly.

Projects

Cindr: Cell Segmentation Software (LTRI Project) | *Python, Flask, React, DeepCell, OpenCV* 2025

- Developed an interactive tool for pathologists to refine ML-driven cell segmentation, reducing manual correction overhead.
- Integrated an AI component to suggest real-time segmentation refinements, assisting in identifying subtle boundary inaccuracies.
- Collaborated with researchers to validate model outputs and iterate based on user feedback and active learning loops.

Battleship Solitaire Solver | *Python, CSP, Backtracking* Nov 2024

- Implemented a Battleship puzzle solver using constraint satisfaction, backtracking, and forward checking algorithms.
- Optimized recursion and variable heuristics for faster puzzle resolution.

Checkers Endgame Engine | *Python, Alpha-Beta Pruning, Game Tree Search* Oct 2024

- Leveraged alpha-beta pruning to develop an efficient AI for endgame checkers scenarios.
- Used heuristic evaluations to enhance move selection and reduce search space.

Face Detector Application | *Python, OpenCV* Jul 2024

- Created a real-time face detection tool using Haar Cascade Classifiers, running efficiently on live webcam feeds.
- Demonstrated proficiency in computer vision and optimized application performance.

Multiplayer Game Using Sockets | *C, Unix Sockets, Concurrent Programming* Mar 2024

- Built a socket-based multiplayer server for a text-based battle game inspired by Pokémon.
- Implemented real-time messaging and turn-based combat for multiple concurrent players.

Unix Shell Implementation | *C, Process Control, IPC* Mar 2024

- Developed a custom Unix shell supporting background processes, piping, and signal handling.
- Demonstrated strong knowledge of system calls and process management.

Interactive Java Game Board | *Java, GUI, Git Flow* Nov – Dec 2023

- Led a 4-person team to build a board game with single/multiplayer modes and interactive audio/visual components.
- Followed SDLC and design patterns, managing sprints and code reviews for efficient merges.

Technical Skills

Languages: Python, Java, C/C++, SQL (Postgres), JavaScript, HTML/CSS, Assembly, Bash/Shell Scripting
Frameworks: React, Node.js, Flask, FastAPI, JUnit, RISC-V, RESTful APIs
Developer Tools: AWS, Git, Virtual Machines (VMs), VS Code, Visual Studio, PyCharm, IntelliJ, Agile, Waterfall
Libraries: pandas, NumPy, Matplotlib, JavaFX
Data Structures: Graphs, Linked Lists, Heaps, Stacks, Queues, Trees, Hash Tables
AI and Algorithms: Alpha-Beta Pruning, A*, Heuristic Search, DFS/BFS, Minimax Algorithm, Game Tree Search, Constraint Satisfaction, Reasoning with Uncertainty
Information Security: Preventing Buffer Overruns, Software Security and Virus Prevention, System Security, Symmetric/Asymmetric Encryption, Hashing, Digital Signatures, Firewalls, VPNs, TCP/IP, Certificates